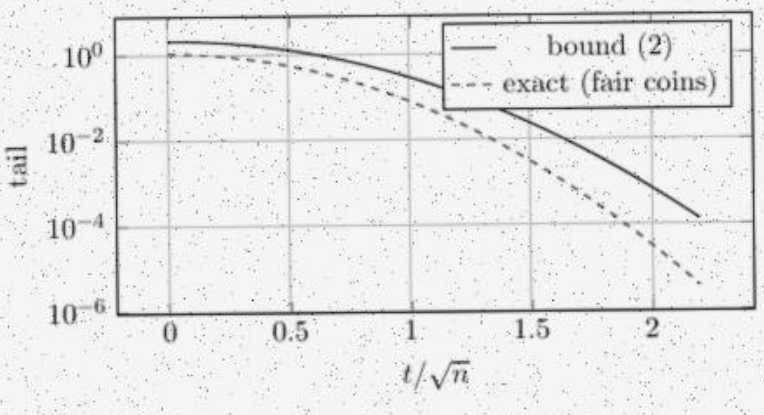


Figure 1: The bound and the exact tail.

n	t	bound	exact
10	3	0.330	0.109
100	10	0.271	0.057
1000	40	0.081	0.012

Table 1: Some values of the bound and of the exact tail.

4 Discussion

The bound of Theorem 2 needs only the range of each variable. A bound with the variance is sharper when the variance is small; see [2]. The code for the figures is at `https://example.com/concentration`. We end with an open question about the case of dependent variables.

References

[1] W. Hoeffding. Probability inequalities for sums of bounded random variables. *J. Amer. Statist. Assoc.*, 58:13–30, 1963.

[2] S. Boucheron, G. Lugosi and P. Massart. *Concentration Inequalities*. Oxford University Press, 2013.